PROJECT DOCUMENT

Mitsue Project

Mitsue Sustainable Energy & AI Data Center

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Project Overview for Consul General Ms. Sandra Pellegrom
Follow-up to King's Day Meeting — April 24, 2026

The Idea in Brief

Repurpose an underused village building — the **Mitsue Taiken Koryukan** is the leading candidate, final choice confirmed in Phase 1 — as a **rural sustainable energy hub** combining:

1. **Native forest restoration** — replacing sugi (cedar) plantations with native broadleaf species; native trees provide natural food for deer, wild boar, and bear, keeping wildlife in the forest and out of crop areas
2. **EV charging infrastructure & community energy resilience** — privately owned solar and EV charging stations at the Koryukan; battery storage subject to feasibility study
3. **A community-owned AI data center** — powered entirely by locally generated renewable energy; a working model demonstrating that technological progress and ecological sustainability are a single coherent system
A single integrated project that addresses ecology, energy, rural revitalization, and digital infrastructure in one place.

Timeframe: 25 Years

The project is planned as a **25-year initiative**, designed to bridge the gap until **localized small-scale fusion power generation** becomes commercially available. Solar and EV charging infrastructure is the practical, sustainable solution for the coming decades — fusion is the long-term horizon.

Why Now — The EV Power Demand

Within roughly **10 years, most cars in Japan will be electric vehicles**. This will require dramatically more **localized electric power generation** — especially in the countryside, where grid extension is expensive and slow.
Rural Japan needs distributed, locally generated power. The Mitsue project fits exactly into this future: a small-scale model that can be replicated across other depopulated villages, providing both EV charging capacity and digital infrastructure.

Project Coordination — A Dedicated NGO

A **non-profit organization (NGO)** will be established to coordinate the project. The NGO will be the neutral hub bringing together a wide range of stakeholders:

Core Stakeholders to Engage

- **Local politicians** — Mitsue village government, Nara Prefecture, national contacts
- **Forestry contractors and managers** — for the harvesting operations
- **Private mountain/forest landowners** — for sourcing sugi from privately owned land
- **Data center operators** — Japanese and international, for technical partnership and operational expertise
- **Business partners** — solar and EV charging suppliers and operators, IT infrastructure providers
- **Communications teams** — for outreach, media, and storytelling
- **Academic and research partners** — universities for forestry, energy, and AI research
- **Citizen and community groups** — local residents, Mitsue community, Safecast network
The NGO model also makes it easier to receive **diverse funding** — government grants, private investment, philanthropic support, and corporate partnerships.

Consultants Already Involved

- **Joi Ito** — President, Chiba Institute of Technology; advisor on technology, innovation, and sustainability
- **Ray Ozzie** — Executive Chair, Blues; advisor and possible participant
- **Evin Zoet** — Co-Representative Director, Transom; advisor
- **Yoshiko Zoet-Suzuki** — Co-Representative Director, Transom; advisor All are already engaged as consultants and potential active participants.

Request to the Consul General

Rob would be honored if Ms. Pellegrom would lend **her name in support** of the project — as a public expression of Dutch interest in innovative, sustainable Dutch-Japanese collaboration in rural Kansai.
This is not a request for funding or formal endorsement, but a goodwill association that signals the project's alignment with Dutch values around sustainability, citizen science, and creative partnerships.

Additional Requirements & Considerations

Legal & Regulatory

- **NGO/NPO incorporation in Japan** — register as a 認定NPO法人 (certified NPO) for tax-deductible donations
- **Building permits** — converting the Koryukan to data use requires zoning and structural approvals
- **Forestry permits** — Japan's Forest Act regulates harvesting; coordination with the Forestry Agency (林野庁)
- **EV charging permits** — grid connection approval, safety inspections for charging equipment

- **FIT/FIP feed-in tariff registration** — for any surplus electricity sold to the grid
- **Data center compliance** — data localization rules, cybersecurity standards, possibly APPI (privacy law)

Technical Infrastructure

- **Solar + EV charging system** — privately owned rooftop solar at the Koryukan powers the data center and EV charging stations. Battery storage to be evaluated in feasibility study. The system is sized for data center and EV demand, not for powering the whole village.
- **High-bandwidth fiber connectivity** — essential for any data center, even small-scale; Mitsue's current fiber capacity needs assessment
- **Blackout resilience** — on-site solar generation enables island-mode operation during grid outages, keeping the data center, EV chargers, and emergency services running; battery storage to be evaluated in feasibility study
- **Cooling strategy** — Japan's humid summers are hard on data centers; could leverage Mitsue's cool mountain climate and water resources
- **Heat recovery** — waste heat from servers can warm greenhouses, fish farms, or community buildings (a real opportunity)
- **Seismic engineering** — the Koryukan must be assessed and likely retrofitted for modern earthquake codes

Financial

- **Phased funding model** — don't try to fund all 25 years at once; secure 3–5 year initial phases
- **Diverse revenue streams** — data center hosting fees, electricity sales, carbon credits, government forestry subsidies, EV charging fees
- **Carbon credit certification** — the forest restoration component could qualify under J-Credit or international schemes

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Pitfalls to Avoid

Political and Community

- **Local resistance is the #1 risk.** Rural Japanese communities can be wary of outside projects, especially involving "AI" or industrial use of village land. Build local trust *before* announcing big plans publicly.
- **Don't bypass village elders and community associations** (自治会). Even if local government approves, social license matters more in rural Japan.
- **Avoid the "tech savior" narrative.** Frame the project as the community's project, supported by outside expertise — not the other way around.

Forestry

- **Sugi monoculture is privately fragmented.** Hundreds of small landowners may need to sign off. Create a simple, fair contract template early.
- **Native forest restoration is slow** (20–50 years to mature). Set realistic ecological expectations — don't oversell biodiversity gains in the first decade.

Technical

- **Data centers depreciate fast.** Hardware is obsolete in 5–7 years. Plan for ongoing capital reinvestment, not a one-time build.
- **Don't over-promise scale.** A "micro" data center is realistic; trying to compete with hyperscalers is not. Position it as edge computing or specialized AI workloads.
- **Battery disposal.** Lithium batteries have a finite cycle life (typically 10–15 years). Plan for recycling and replacement costs. Get proper lifecycle accounting from day one.

Organizational

- **NGO governance.** Boards can become dysfunctional. Set clear bylaws, term limits, and conflict-of-interest policies up front.
- **Founder dependency.** Don't let the project depend entirely on one or two people. Build redundancy in leadership.
- **Mission drift.** With many stakeholders, projects can lose focus. Document a clear charter and revisit it annually.

Additional Ideas to Strengthen the Project

Synergies to Add

- **Educational component** — partner with universities for student internships and research projects; the Koryukan setting is symbolic and practical
- **Citizen science integration** — Safecast environmental monitoring (radiation, air quality, forest health) deployed across the project area as a living lab
- **Open data commitment** — publish energy production, forest growth, and environmental data publicly (aligns with your Safecast philosophy and Dutch open-government values)
- **Server waste heat loop** — data center heat can still warm community spaces or a small greenhouse, closing a useful loop without requiring a separate thermal plant
- **EV charging network** — install charging stations in Mitsue and neighboring villages from year one, creating immediate visible value to residents

Strategic Positioning

- **Official policy tailwind (de-risking).** Mitsue Village has its own MoE-funded **"Plan for Maximum Introduction of Renewable Energy"** (御杖村再エネ導入最大化計画, Jan 2025). The project is positioned as the operating vehicle for that official plan, which puts it on a named national funding ladder: the village's completed planning grant makes it eligible for the 地域脱炭素移行・再エネ推進交付金 (2/3–3/4 subsidy on solar/battery/EV, paid to the village). For the consulate this signals the project is embedded in official Japanese policy, not speculative.
- **Make it a model, not a one-off.** Document everything as an open playbook so other Japanese villages can replicate it. This attracts government and philanthropic interest faster.
- **Connect to Expo 2025 legacy.** Osaka Expo's sustainability themes are still alive politically — frame the project as carrying that legacy forward into Kansai's rural areas.
- **Dutch-Japanese branding.** Position as a flagship of Dutch sustainable innovation in Japan, especially around water management, circular economy, and smart agriculture (areas where the Netherlands genuinely leads globally).
- **Engage Dutch corporates already in Japan** — companies like Philips, ASML, Heineken, and others may be interested in CSR partnership.

Communications Strategy

- **Start with a strong founding story** — Mitsue, Safecast, citizen science, the Mitsue Taiken Koryukan. The narrative is already compelling; tell it well.
- **Bilingual from day one** — Japanese-language credibility is essential for local trust; English is essential for international partners.
- **Modest, credible milestones** — under-promise, over-deliver. Rural projects in Japan that hype too early lose credibility quickly.

Project Value & Returns

Qualitative Impact (Available from Day One)

- **Local employment**: 8–15 direct jobs in forestry, facility operations, and EV infrastructure by Year 5
- **Income for landowners**: Fair compensation streams to private mountain owners
- **Energy resilience**: Distributed generation reducing dependence on imports
- **EV infrastructure**: Visible local benefit from the start
- **Replicable model**: Documented playbook for other depopulating villages
- **SDG alignment**: Direct progress on goals 7, 11, 12, 13, 15, 17

Indicative Financial Framework

Phase 1 feasibility studies will produce vetted figures. Early-stage estimates suggest:

Metric	**Year 5 (illustrative)**	**Year 10 (illustrative)**
Annual revenue	¥28–67M	¥74–166M
Operating costs	~60% of revenue	~55% of revenue
Net annual surplus	¥10–32M	¥30–75M
Capital invested by Year 5	¥200–290M	—
Approximate payback period	10–18 years	—

Cost Displacement

- Mitsue currently imports ~~100% of its electricity (~~¥40–60M annually leaves the local economy)
- The Koryukan costs the village ¥3–8M/year in upkeep with no return
- Untended sugi plantations carry ecological and physical risk costs not currently priced

The project converts these standing costs into local value.

These figures are illustrative ranges, not forecasts. Phase 1 feasibility studies — forestry, energy systems, building condition, and connectivity — will replace these with vetted numbers within 9 months of project launch.

Suggested Next Steps

1. **Form a small founding team** (3–5 people) and draft NGO bylaws
2. **Secure a written letter of interest** from Mitsue village government before public announcements
3. **Map the privately owned forest plots** and identify the most willing landowners
4. **Commission a feasibility study** — energy production, fiber connectivity, building condition
5. **Build a simple project website** with bilingual content, founding members, advisory board
6. **Apply for initial seed funding** — government rural revitalization grants (地方創生) are a natural starting point

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